MyLearna
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MATHEMATICS
PROBABILITY AND CHANCE
Judge Fairness and Likelihood More Precisely

Name: ________________

Date: ________________

Step 8

Upper Elementary (Grades 4–5)

Today I am learning to judge fairness and likelihood more precisely.

1. Is It Fair?

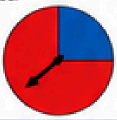
Decide if each game is fair. Write Fair or Not Fair.
Explain your answer using fractions or percentages.

a) A bag has 2 red counters and 2 blue. Your pick one counter.

Fair or Not Fair: ________

Why? ________

b) A spinner has 3 red sections and 1 blue section. You spin once.



Fair or Not Fair: ________

Why? ________

c) A number cube has the numbers 1–6. You roll it once and try to get an even number.

1 2 3 4 5 6

Fair or Not Fair: ________

Why? ________

2. Compare Likelihoods

Compare the likelihood of each event.
Write >, < or =.

a) Rolling a 6 on a number cube ◯ Rolling an even number

b) Spinning red on this spinner ◯ Spinning blue on this spinner

c) Picking a red counter from this bag ◯ Picking a blue counter from this bag

d) Getting heads on a coin toss ◯ Getting tails on a coin toss

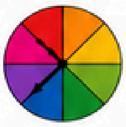
e) Rolling a number less than 3 on a number cube ◯ Rolling a number greater than 3 on a number cube

f) Picking a green counter from this set of counters ◯ Picking a blue counter from this set of counters

3. Order the Chances

Order the events from least likely to most likely.
Write 1 for least likely and 4 for most likely.

a) A spinner with 8 equal sections.



land on red	land on blue	land on a primary colour	land on a warm colour
☐	☐	☐	☐

b) A bag with 8 counters: 3 red, 3 blue, 1 green, 1 yellow.

pick red	pick blue	pick green	pick yellow
☐	☐	☐	☐

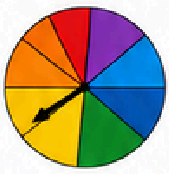
4. Two Events – Which Is More Likely?

For each pair of events, decide which is more likely.
Tick (✓) your answer and explain why.

	Event A	or	Event B	More Likely A	More Likely B	Why?
a)	Rolling a 2	or	Rolling a 1 or 2	☐	☐	
b)	Spinning yellow on this spinner	or	Spinning green on this spinner	☐	☐	
c)	Picking a red counter from this bag	or	Picking a blue counter from this bag	☐	☐	
d)	Getting tails on two coin tosses	or	Getting heads on two coin tosses	☐	☐	

5. Probabilities as Fractions, Decimals and Percentages

A spinner has 10 equal sections as shown.



Event	Fraction	Decimal	Percentage
land on red			
land on blue			
land on yellow			
land on green			
land on purple			

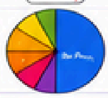
6. Real-Life Fairness

Look at these situations. Decide if each is fair.
Write Fair or Not Fair. Explain your answer.

a) Two players each roll a number cube. The higher number wins.
Is it fair? ________ Why? ________

b) A game where you draw a card from a standard deck. If you get a heart, you win a prize.
Is it fair? ________ Why? ________

c) A prize wheel with 1 very large section for "No Prize" and 5 small sections for prizes.
Is it fair? ________ Why? ________



7. Plan a Fair Game

Design a game that is fair.
Use counters, a spinner or a number cube.
Show your game and explain why it is fair.

My Game:

Why it is fair: ________

8. Think and Explain

a) Why is it important to use fractions, decimals or percentages to describe chance? ________

b) How can knowing the likelihood of outcomes help us make better choices? ________

c) Give an example of a time when you used chance to make a decision. ________

I can...

- ☐ judge if a game is fair.
- ☐ compare likelihoods precisely.
- ☐ order events from least likely to most likely.
- ☐ use fractions, decimals and percentages.
- ☐ design fair games and explain my reasoning.

Chance Words

fair, unfair, likely, unlikely, more likely, less likely, equally likely, chance, probability, outcome

impossible	unlikely	equally likely	likely	certain
0	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{3}{4}$	1
0%	25%	50%	75%	100%

Great job! You are using maths to make fair and smart decisions!